

Robot Hardware Evidence Report

Generated: 2026-07-04T21:19:18

Server state: dry_run=False, motion_enabled=True, protocol=feetech_sts, port=COM11

Scenario	Command result	Servo deltas	Mean diff	Changed pixels
01_scan	scan found ids: [2, 3, 4, 5]	base: 0 elbow: 0 gripper: None shoulder: 0 wrist: 0	2.28	0.07%
02_shoulder_positive	ST/SC servo 3: 1897 -> target 1957 -> readback 1904 in 600ms (ack)	base: 0 elbow: 0 gripper: None shoulder: 8 wrist: 0	12.35	15.85%
03_shoulder_negative	ST/SC servo 3: 1905 -> target 1900 -> readback 1898 in 600ms (ack)	base: 0 elbow: 0 gripper: None shoulder: -8 wrist: 0	11.99	15.44%
04_shoulder_negative	ST/SC servo 3: 1897 -> target 1900 -> readback 1897 in 600ms (ack)	base: 0 elbow: 0 gripper: None shoulder: 0 wrist: 0	2.29	0.07%
05_elbow_positive	ST/SC servo 4: 638 -> target 698 -> readback 694 in 600ms (ack)	base: 0 elbow: 56 gripper: None shoulder: 0 wrist: 0	26.74	35.31%
06_wrist_negative	ST/SC servo 5: 3364 -> target 3304 -> readback 3307 in 600ms (ack)	base: 0 elbow: 0 gripper: None shoulder: 0 wrist: -57	26.87	31.04%
07_wrist_positive	ST/SC servo 5: 3307 -> target 3367 -> readback 3363 in 600ms (ack)	base: 0 elbow: 0 gripper: None shoulder: 0 wrist: 56	26.85	30.99%

01_scan

Scan the transparent ST/SC serial bus for responding servo IDs.

Action JSON: {"type": "scan", "start_id": 1, "end_id": 12}

App result: scan found ids: [2, 3, 4, 5]

Servo before: {"base": 151, "elbow": 638, "gripper": null, "shoulder": 1897, "wrist": 3364}

Servo after: {"base": 151, "elbow": 638, "gripper": null, "shoulder": 1897, "wrist": 3364}

Servo delta: {"base": 0, "elbow": 0, "gripper": null, "shoulder": 0, "wrist": 0}

Movement score: mean_abs_diff=2.28, changed_pixels=0.07%

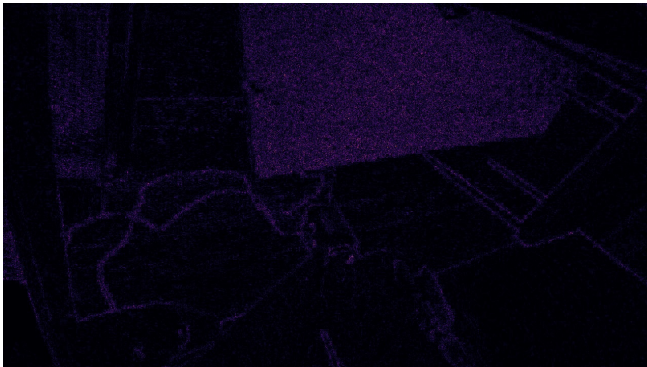
Before



After



Difference heatmap



02_shoulder_positive

Move shoulder joint by +60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"shoulder","delta":60}

App result: ST/SC servo 3: 1897 -> target 1957 -> readback 1904 in 600ms (ack)

Servo before: {"base":151,"elbow":638,"gripper":null,"shoulder":1897,"wrist":3364}

Servo after: {"base":151,"elbow":638,"gripper":null,"shoulder":1905,"wrist":3364}

Servo delta: {"base":0,"elbow":0,"gripper":null,"shoulder":8,"wrist":0}

Movement score: mean_abs_diff=12.35, changed_pixels=15.85%

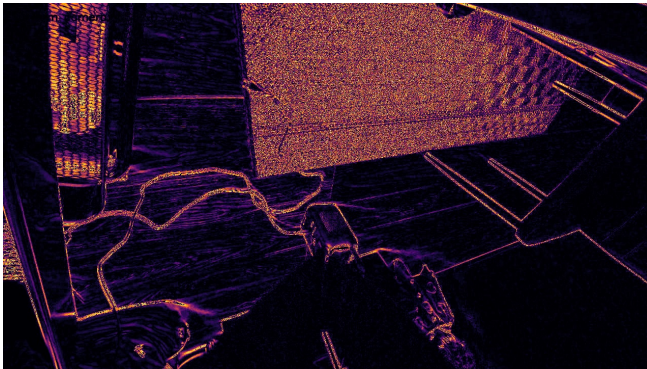
Before



After



Difference heatmap



03_shoulder_negative

Move shoulder joint by -60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"shoulder","delta":-60}

App result: ST/SC servo 3: 1905 -> target 1900 -> readback 1898 in 600ms (ack)

Servo before: {"base":151,"elbow":638,"gripper":null,"shoulder":1905,"wrist":3364}

Servo after: {"base":151,"elbow":638,"gripper":null,"shoulder":1897,"wrist":3364}

Servo delta: {"base":0,"elbow":0,"gripper":null,"shoulder":-8,"wrist":0}

Movement score: mean_abs_diff=11.99, changed_pixels=15.44%

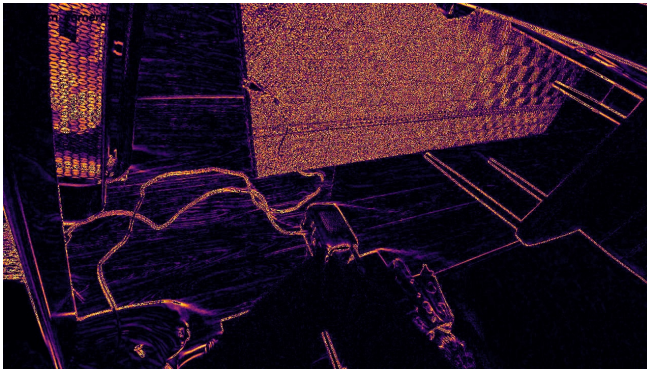
Before



After



Difference heatmap



04_shoulder_negative

Move shoulder joint by -60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"shoulder","delta":-60}

App result: ST/SC servo 3: 1897 -> target 1900 -> readback 1897 in 600ms (ack)

Servo before: {"base":151,"elbow":638,"gripper":null,"shoulder":1897,"wrist":3364}

Servo after: {"base":151,"elbow":638,"gripper":null,"shoulder":1897,"wrist":3364}

Servo delta: {"base":0,"elbow":0,"gripper":null,"shoulder":0,"wrist":0}

Movement score: mean_abs_diff=2.29, changed_pixels=0.07%

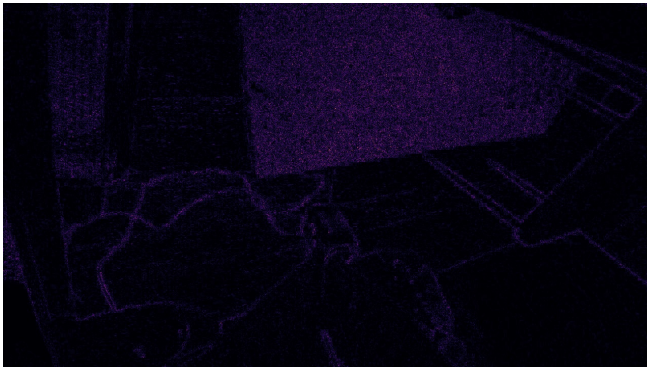
Before



After



Difference heatmap



05_elbow_positive

Move elbow joint by +60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"elbow","delta":60}

App result: ST/SC servo 4: 638 -> target 698 -> readback 694 in 600ms (ack)

Servo before: {"base":151,"elbow":638,"gripper":null,"shoulder":1897,"wrist":3364}

Servo after: {"base":151,"elbow":694,"gripper":null,"shoulder":1897,"wrist":3364}

Servo delta: {"base":0,"elbow":56,"gripper":null,"shoulder":0,"wrist":0}

Movement score: mean_abs_diff=26.74, changed_pixels=35.31%

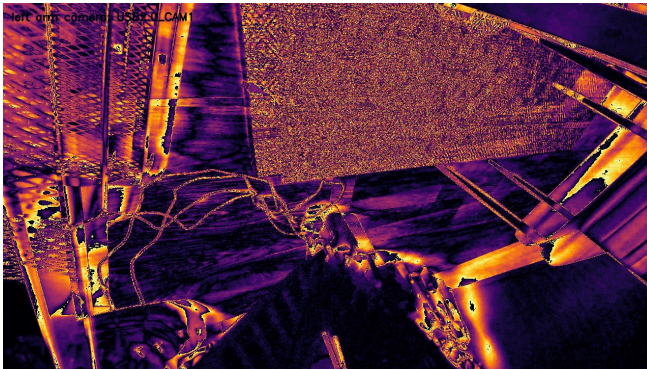
Before



After



Difference heatmap



06_wrist_negative

Move wrist joint by -60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"wrist","delta":-60}

App result: ST/SC servo 5: 3364 -> target 3304 -> readback 3307 in 600ms (ack)

Servo before: {"base":151,"elbow":694,"gripper":null,"shoulder":1897,"wrist":3364}

Servo after: {"base":151,"elbow":694,"gripper":null,"shoulder":1897,"wrist":3307}

Servo delta: {"base":0,"elbow":0,"gripper":null,"shoulder":0,"wrist":-57}

Movement score: mean_abs_diff=26.87, changed_pixels=31.04%

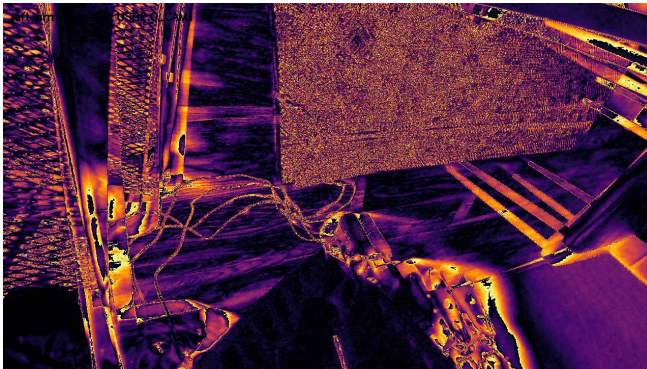
Before



After



Difference heatmap



07_wrist_positive

Move wrist joint by +60 ST/SMS position units from live readback.

Action JSON: {"type":"nudge_joint","joint":"wrist","delta":60}

App result: ST/SC servo 5: 3307 -> target 3367 -> readback 3363 in 600ms (ack)

Servo before: {"base":151,"elbow":694,"gripper":null,"shoulder":1897,"wrist":3307}

Servo after: {"base":151,"elbow":694,"gripper":null,"shoulder":1897,"wrist":3363}

Servo delta: {"base":0,"elbow":0,"gripper":null,"shoulder":0,"wrist":56}

Movement score: mean_abs_diff=26.85, changed_pixels=30.99%

Before



After



Difference heatmap

